

Chapter 6, Problem 2P

Problem

A $50\text{-}\mu\text{F}$ capacitor has energy $w(t) = 10 \cos^2 377t$ J. Determine the current through the capacitor.

Step-by-step solution

Step 1 of 1

The capacitance of a capacitor, $C = 50\text{ }\mu\text{F}$.

The energy stored in capacitor is,

$$w(t) = 10 \cos^2 377t \text{ J}$$

Write the formula for energy stored in the capacitor.

$$w(t) = \frac{1}{2} C v^2(t)$$

Substitute 50×10^{-6} for C and $10 \cos^2 377t$ for $v(t)$.

$$10 \cos^2 377t = \left(\frac{1}{2}\right) (50 \times 10^{-6}) v^2(t)$$

$$20 \cos^2 377t = (50 \times 10^{-6}) v^2(t)$$

$$v^2(t) = 400000 \cos^2 377t$$

$$v(t) = \sqrt{4 \times 10^5 \cos^2 377t}$$

$$v(t) = 632.45 \cos 377t \text{ V}$$

Determine the current through the capacitor using the following formula.

$$i_c = C \frac{dv(t)}{dt}$$

$$\begin{aligned} i_c &= 50 \times 10^{-6} \times \frac{d}{dt} (632.45 \cos 377t) \\ &= (50 \times 10^{-6}) (632.45) 377 (-\sin 377t) \\ &= -11.92 \sin 377t \text{ A} \end{aligned}$$

Thus, the current through the capacitor is $\boxed{-11.92 \sin 377t \text{ A}}$.